

Problem: From the following table calculate UN-Age-Sex accuracy index and comment.

	Age Group	Population	
		Male	Female
	0-4	1022001	965554
	5-9	964951	916269
	10-14	841146	792179
	15-19	531428	503091
	20-24	318826	470133
	25-29	445106	429185
	30-34	423287	377645
	35-39	400610	325684
	40-44	336099	265734
	45-49	270331	209086
	50-54	236986	186234
	55-59	163501	143373
	60-64	108910	105412
	65-69	67351	74716
	70-74	36557	48300

Solution: UN-Age-Sex accuracy index has three components;

(a) Average sex ratio score (S):

This score is obtained by first calculating the sex ratio at each group. Then successive differences irrespective of sign are added and averaged.

$$\text{Age-specific sex ratio} = \frac{{}_5P_x^m}{{}_5P_x^f} \times 100$$

where, ${}_5P_x^m$ = males aged x to $x+5$ and

${}_5P_x^f$ = females aged x to $x+5$

(b) Average male age ratio score (M):

For each age group for males, calculate the age ratios computed as

$$\text{Age ratio} = \frac{{}_5P_x}{\frac{1}{2}({}_5P_{x-5} + {}_5P_{x+5})} \times 100$$

The deviations from unity irrespective of sign are added and averaged (M).

(c) Average female age ratio score (F):

For each age group for females, the age ratios are calculated using the same formulae as for males.

$$\text{Age-Ratio} = \frac{{}_5P_x}{\frac{1}{2}({}_5P_{x-5} + {}_5P_{x+5})} \times 100$$

The deviations from unity irrespective of sign are added and averaged (F).

The index is then computed as :

$$\text{UNAI} = 3(S) + M + F$$

The reported age-sex data for a given population is presumed to be:

1. "accurate" if the age sex accuracy index is between 0 and 19.9,
2. "inaccurate" if the index is between 20 and 39.9, and
3. "highly inaccurate" if the index is above 40.

The relevant calculation is given in the following table by using MS Excel.

Age Group	Population		Sex Ratio	Successive Difference
	Male	Female		
0-4	1022001	965557	105.9461	
5-9	964956	916239	105.3136	0.5324
10-14	841146	799179	105.2466	0.0649
15-19	531428	503091	105.6326	0.3839
20-24	318826	470153	67.8132	37.8193
25-29	445106	429185	103.7196	35.8963
30-34	423287	377645	112.086	8.3763
35-39	400660	325684	123.0211	10.9351
40-44	336097	265734	126.4796	3.4583
45-49	270331	209086	129.2918	2.8122
50-54	236988	186234	127.2528	2.0389
55-59	163501	143373	114.0389	13.2139
60-64	108910	105412	103.3184	10.7204
64-69	67351	74716	90.1426	13.1757
70-74	36587	48300	75.7494	14.3931
Total				153.8215
Average				S = 10.9872

	Male Age Ratio	Male Deviation from 100	Female Age Ratio	Female Deviation from 100
	103.5834	3.5834	103.8410	3.8410
	112.4238	12.4238	112.6139	12.6139
	91.6277	8.3722	79.2673	20.7326
	65.2974	34.7025	100.8613	0.8613
	119.9563	19.9563	101.2469	1.2469
	100.0955	0.0955	100.0557	0.0557
	105.5220	5.5220	106.2417	1.2417
	100.1798	0.1798	99.3825	0.6174
	94.3420	5.6579	92.3224	7.4775
	109.2533	9.2533	105.6769	5.6769
	94.5371	5.4628	98.3198	1.6801
	94.3548	5.6451	96.6667	3.3312
	92.5806	7.4192	97.0155	2.7844
	1283.7543	118.2747	1288.9146	62.1612
	98.7503	M = 9.0980	99.14267	F = 4.7816

From the table,

$$S = 10.9872$$

$$M = 9.0980$$

$$F = 4.7816$$

$$\therefore \text{UNAI} = 3(S) + M + F$$

$$= 3 \times 10.9872 + 9.0980 + 4.7816$$

$$= 46.8415$$

Comment: The value of UNAI is above 40. It indicates that the reported age-sex data for a given population is highly inaccurate. There is likely significant age misreporting. Common issues age heaping (clustering of ages ending in 0 or 5), digit preference etc. It suggests that respondents may not have reported their exact ages correctly, or enumerators recorded them inaccurately.

The dataset should be used with caution. Demographic analysis (like fertility, mortality etc.) may be biased. Data adjustment or smoothing techniques must be needed before serious use.

Problem: Using age distribution data for a West African country (1960), determine the extent of heaping on

- Digit '0' and '5'
- Digit '0' only
- Digit '5' only, using Whipple's index and interpret the result.

Age	Population	Age	Population
23	38687	45	44654
24	51239	46-49	77102
25	77141	50	52024
26-29	201450	51-54	44751
30	110379	55	14024
31-34	132196	56-59	45283
35	64091	60	38377
36-39	134140	61	4779
40	81515	62	7866
41-44	84422	Total	1304110

Solution: Whipple's index gives the relative preference for digit '0' and '5' while reporting age in the interval 23 to 62 years. It varies between 100 to 500.

If the heaping on terminal digit "0" and "5" is measured

$$\text{WI or Index} = \frac{\sum (P_{25} + P_{30} + \dots + P_{55} + P_{60})}{\frac{1}{5} \sum (P_{23} + P_{24} + \dots + P_{61} + P_{62})} \times 100$$

where, P_x = the no. of persons reporting their age as x years

If the heaping on terminal digit "0" is measured

$$WI = \frac{P_{30} + P_{40} + P_{50} + P_{60}}{\frac{1}{10} \sum (P_{23} + P_{24} + \dots + P_{61} + P_{62})} \times 100$$

If the heaping on terminal digit "5" only is measured

$$WI = \frac{P_{35} + P_{45} + P_{55} + P_{65}}{\frac{1}{10} \sum (P_{23} + P_{24} + \dots + P_{61} + P_{62})} \times 100$$

The index can be grouped in the following categories:

* If $WI = 100$, data has no heaping

* If $WI = 500$, data has only '0' and '5' digit heaping

* If $WI < 105$, data is very accurate

* If $WI = 105 - 110$, data is relatively accurate

* If $WI = 110 - 125$, data is good

* If $WI = 125 - 175$, data is bad

* If $WI > 175$, data is very bad

The relevant calculation given in the following table by using Ms Excel.

Age	Population	Age	Population
25	77141	25	77141
30	110379	35	64091
35	64091	45	44654
40	81515	55	14024
45	44654	Total	199910
50	52024	30	110379
55	14024	40	81515
60	38377	50	52024
		60	38377
Total	482205	Total	282295

Given that, total population = $\Sigma(P_{25} + P_{30} + \dots + P_{55} + P_{60})$
 $= 1304110$

From the table,

$$\Sigma(P_{25} + P_{30} + \dots + P_{55} + P_{60}) = 482205$$

$$P_{25} + P_{35} + P_{45} + P_{55} = 199910$$

$$P_{30} + P_{40} + P_{50} + P_{60} = 282295$$

(a) Age heaping on '0' and 5,

$$WI(0,5) = \frac{482205}{\frac{1}{5} \times 1304110} \times 100$$

$$= 184.8790$$

(b) Age heaping on digit '0' only,

$$WI(0) = \frac{282295}{\frac{1}{10} \times 1304110} \times 100$$

$$= 216.4656$$

(c) Age heaping on digit '5' only,

$$WI(5) = \frac{199910}{\frac{1}{10} \times 1304110} \times 100$$
$$= 153.2923$$

Comment:

Age Heaping	WI	Comment
Digit '0' and '5'	184.8790	very bad
Digit '0' only	216.4456	very bad
Digit '5' only	153.2923	bad

For question (a), it indicates very strong age heaping on ages ending in 0 and 5. For (b), respondents show a very strong preference for ages ending in 0. Whipple's index for digit '5' only is 153.2923, this also noticeable heaping, though less severe than for digit 0.

The age data suffer from substantial digit preference especially for ages ending in 0. Respondents were much more likely to report rounded ages (particularly multiples of 10), indicating very poor age reporting accuracy for that West African country. The heaping on digit 0 is stronger than on digit 5.

Problem: Use the Myer's blended index to assess the quality of age data given below :-

Digit	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
0	386	417	307	190	89	80	50	40	10
1	133	93	90	89	41	35	33	10	2
2	341	227	100	40	30	25	15	6	4
3	223	160	90	46	28	22	8	5	1
4	201	138	50	38	25	10	8	5	3
5	298	238	201	154	145	55	25	2	3
6	198	105	70	49	31	18	12	6	4
7	166	86	50	45	25	9	5	4	2
8	255	100	90	40	28	10	5	5	2
9	132	85	40	20	8	6	2	2	2

Solution: Myer's blended index procedure for computations:

1. Sum all the populations ending in each digit over the whole range i.e. 10-99.

2. Sum figures between ages 20-99.

3. Multiply the sums in (1) by coefficients; 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10.

4. Multiply the sums in (2) by coefficients from 9 descending to 0 i.e. 9, 8, 7, 6, 5, 4, 3, 2, 1, 0.

5. Add the product of (3) and (4), to obtain the blended sum.

6. Add up the blended sum in (5).

7. Find the percent (%) of the total blended sum of different digit ends.

8. Take the deviations of each % in (7) from 10. This result indicates the extent of concentration or avoidance of a particular digit.

						Blended	% :	Absolute
Sum1	Weight1	Product1	Sum2	Weight2	Product2	sum	Blended	Deviation
						(P ₁ +P ₂)	Population	from 10
1569	1	1569	1183	9	10647	12216	21.6945	11.6945
526	2	1052	393	8	3144	4196	7.4517	2.5482
788	3	2364	447	7	3129	5493	9.7551	0.2448
583	4	2332	311	6	2160	4492	7.9774	2.0225
478	5	2390	277	5	1385	3775	6.7040	3.2959
1134	6	6804	536	11	3344	10148	18.0219	8.0219
493	7	3451	275	3	885	4336	7.7003	2.2996
392	8	3136	226	2	452	3588	6.3719	3.6280
1535	9	4815	200	1	280	5095	9.0482	0.9517
297	10	2970	15	0	0	2970	5.2744	4.7255
Total						56309	100	39.4331

Interpretation of Myer's blended index:

MI	Interpretation	Quality of Age Data
0	No age heaping	Perfect quality
0-5	very little age preference	Very good
5-10	slight age heaping	Good
10-20	Moderate age heaping	acceptable
20-30	High age heaping	Poor
Above 30	very severe age heaping	Very Poor

$$\text{Myer's Blended Index} = \frac{1}{2} \sum |P_i - 10|$$

From table, $\sum |P_i - 10| = 39.4331$,
where, $P_i = \% \text{ blended population}$

$$\therefore \text{Myer's Blended Index} = \frac{1}{2} \times 39.4331 = 19.7165$$

Comment: The age data shows considerable age heaping. Digits 0 and 5 are strongly preferred, especially digit 0 with 21.73% and digit 5 with 18.04%, instead of the expected 10% each.

Our calculated Myer's index is 19.77, indicates moderate to high age heaping. The age data is fair or acceptable quality but not fully reliable age data. Strong preference for ages ending in 0 and 5.

Problem: The following table gives the census population of Bangladesh for the years 2011 and 2021.

	Age Group	Population	
		2011	2021
	0-4	195300	200000
	5-9	180700	185000
	10-14	165400	170090
	15-19	155200	165000
	20-24	148400	150400
	25-29	143300	145000
	30-34	140100	140600
	35-39	137200	135800
	40-44	130200	130800
	45-49	123700	125000
	50-54	114700	115600
	55-59	110800	113400
	60-64	93750	96500
	65-69	84782	85700
	70-74	73856	75300
	75-79	56500	60000
	80+	44000	45500

(a) Draw a population pyramid

(b) Compute intercensal growth rate using different growth functions.

(c) Find the time to double the population size of 2011.

(d) Calculate the projected population for the years 2041 and 2050.

(e) Make an overall comment

(a)

Solution: For drawing population pyramid the relevant calculation given in the following table:

Age Group	Population		% Population	
	2011	2021	2011	2021
0-4	195300	200000	-9.3093	9.3471
5-9	180700	185000	-8.6134	8.6461
10-14	165400	170090	-7.8841	7.9492
15-19	155200	165000	-7.3979	7.7113
20-24	148400	150400	-7.0737	7.0290
25-29	143300	145200	-6.8306	6.7766
30-34	140100	140500	-6.6781	6.5710
35-39	137200	135700	-6.5399	6.3467
40-44	130200	130300	-6.2062	6.1130
45-49	123700	125000	-5.8964	5.8419
50-54	114700	115600	-5.4674	5.4026
55-59	110800	113400	-5.2815	5.2998
60-64	93750	96500	-4.4687	4.5099
65-69	84782	85700	-4.0413	4.0052
70-74	73856	75300	-3.5204	3.5192
75-79	56500	60000	-2.6931	2.8041
80+	44000	45500	-2.0973	2.1264
Total	2097888	2139690	-100	100

graph:

Comment: The graph seems to represent an expansive population pyramid. It has a wide base—meaning there is a large proportion of children and young people. The pyramid gradually narrows toward the top—showing that the number of older people is smaller. This shape usually indicates high birth rates, rapid population growth and lower life expectancy compared to developed populations. These types of pyramids are seen in developing countries.

(b) For linear growth function,

$$r = \frac{P_t - P_0}{P_0 t}$$

where, P_t = Population at the end year

P_0 = Population at the current/last year

t = time period

r = growth rate

From the table, $P_t = 2139690$

$P_0 = 2097888$

$t = 10$ years

$$\begin{aligned} \therefore r &= \frac{2139690 - 2097888}{2097888 \times 10} \\ &= 0.00199 \end{aligned}$$

For geometric growth function,

$$\begin{aligned} r &= \left(\frac{P_t}{P_0} \right)^{\frac{1}{t}} - 1 \\ &= \left(\frac{2139690}{2097888} \right)^{\frac{1}{10}} - 1 \\ &= 0.00197 \end{aligned}$$

For exponential growth function,

$$r = \frac{1}{t} \ln\left(\frac{P_t}{P_0}\right)$$
$$= \frac{1}{10} \ln\left(\frac{2139694}{2097888}\right)$$
$$= 0.00197$$

(c) Calculating the time to double the population size of 2011

For linear growth function,

$$t = \frac{1}{r} = \frac{1}{0.00197} = 511.8690 \text{ Years}$$

For geometric growth function,

$$t = \frac{\ln 2}{\ln(1+r)} = \frac{\ln 2}{\ln(1+0.00197)} = 351.3193 \text{ Years}$$

For exponential growth function,

$$t = \frac{\ln 2}{r} = \frac{\ln 2}{0.00197} = 351.3193 \text{ Years}$$

(d) Calculating the projected population for the years 2041 and 2050;

For linear growth function,

$$P_{30} = P_0(1+rt) = 2097888(1+0.00197 \times 30) = 2223294$$

$$P_{39} = 2097888(1+0.00197 \times 39) = 2260916$$

For geometric growth function,

$$P_{30} = P_0 (1+r)^t = 2097888 (1+0.00197)^{30} = 222809$$

$$P_{39} = 2097888 (1+0.00197)^{39} = 2265686$$

For exponential growth function,

$$P_{30} = P_0 e^{rt} = 2097888 e^{0.00197 \times 30} = 2225809$$

$$P_{39} = 2097888 \times e^{0.00197 \times 39} = 2265686$$

(e) Overall comments: At first we draw population pyramid. It seems like error in population pyramid. After that we calculate interannual growth rate for linear, geometric and exponential growth functions. The values are 0.199%, 0.197% and 0.197% respectively. In question no. (c), we compute the time to double the population size of 2011 for the above three cases, 501.8630 years, 351.3193 years and 351.3193 years are the values of that growth functions respectively. In this question, the values for geometric and exponential growth function are same. But for linear growth functions, it is relatively large. It seems inaccurate result. So, linear growth functions gives inaccurate result for predicting time. In next question, we calculate the projected population for the years 2041 and 2050 respectively for each growth function.